

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285666

Kind Code

A1

Publication Date

September 11, 2025

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MEMORY SYSTEM

Abstract

According to one embodiment, a memory system includes a semiconductor memory, a controller, and a first circuit. The semiconductor memory includes a nonvolatile memory cell. The controller is configured to cause the semiconductor memory to execute first and second write operations. The first write operation writes a first bit into the memory cell. The second write operation writes first data based on the first bit and a second bit into the memory cell. The first circuit checks whether or not the first bit includes a bit error. The controller is configured to cause the semiconductor memory to execute, in the second write operation, writing of the first data including the second bit and a third bit obtained by correcting the bit error of the first bit, in a case that the first bit includes the bit error.

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Appl. No.: 19/220765

Filed: May 28, 2025

Foreign Application Priority Data

JP	2022-107105	Jul. 01, 2022
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Related U.S. Application Data

parent US continuation 18068914 20221220 parent-grant-document US 12347518 child US 19220765

Publication Classification

Int. Cl.: G11C7/10 (20060101); G11C29/52 (20060101)

U.S. Cl.:

CPC G11C7/1096 (20130101); G11C7/1069 (20130101); G11C29/52 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of and claims benefit under 35 U.S.C. § 120 to U.S. application Ser. No. 18/068, 914, filed Dec. 20, 2022, which is based upon and claims the benefit of priority under 35 U.S.C. § 119 from Japanese Patent Application No. 2022-107105, filed Jul. 1, 2022, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a memory system.

BACKGROUND

[0003] A memory system including a semiconductor memory and a memory controller configured to control the semiconductor memory is known.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a block diagram illustrating an example of an information processing system including a memory system according to a first embodiment.

[0005] FIG. 2 is a block diagram illustrating an example of the configuration of a semiconductor memory included in the memory system according to the first embodiment.

[0006] FIG. 3 is a circuit diagram of a memory cell array inside the semiconductor memory included in the memory system according to the first embodiment.

[0007] FIG. 4 is a diagram illustrating an example of a relationship between threshold voltage distributions of memory cell transistors and data stored therein, which memory cell transistors are included in the memory system according to the first embodiment.

[0008] FIG. 5 is a sequence diagram illustrating an example of a first write operation in the memory system according to the first embodiment.

[0009] FIG. 6 is a diagram illustrating changes in threshold voltage distributions of the memory cell transistors caused by the first write operation in the memory system according to the first embodiment.

[0010] FIG. 7 is a sequence diagram illustrating an example of a second write operation in the memory system according to the first embodiment.

[0011] FIG. 8 is a diagram illustrating changes in threshold voltage distributions of the memory cell transistors caused by the second write operation in the memory system according to the first embodiment.

[0012] FIG. 9 is a diagram illustrating an example of the order of executing the first write operation and the second write operation in the memory system according to the first embodiment.

[0013] FIG. 10 is a conceptual diagram illustrating an example of a write management table of the memory system according to the first embodiment.

[0014] FIG. 11 is a conceptual diagram illustrating an example of a write history table of the memory system according to the first embodiment.

[0015] FIG. 12 is a flowchart illustrating an example of a data write operation in the memory system according to the first embodiment.

[0016] FIG. 13 is a flowchart illustrating an example of a padding determination operation in the memory system according to the first embodiment.

[0017] FIG. 14 is a diagram illustrating an example of a data size for use in a padding operation in the memory system according to the first embodiment.

[0018] FIG. 15 is a diagram illustrating another example of a data size for use in the padding operation in the memory system according to the first embodiment.

[0019] FIG. 16 is a flowchart illustrating an example of the padding operation in the memory system according to the first embodiment.

[0020] FIG. 17 is a block diagram illustrating an example of an information processing system including a memory system according to a second embodiment.

[0021] FIG. 18 is a block diagram illustrating an example of the configuration of a semiconductor memory included in the memory system according to the second embodiment.

[0022] FIG. 19 is a flowchart illustrating an example of a data write operation in the memory system according to the second embodiment.

[0023] FIG. 20 is a flowchart illustrating an example of a recovery operation in the memory system according to the second embodiment.

[0024] FIG. 21 is a flowchart illustrating an example of a recovery operation in a memory system according to a first modification of the second embodiment.

[0025] FIG. 22 is a flowchart illustrating an example of a data write operation in a memory system according to a second modification of the second embodiment.

[0026] FIG. 23 is a block diagram illustrating an example of an information processing system including a memory system according to a third modification of the second embodiment.

[0027] FIG. 24 is a conceptual diagram illustrating an example of a number-of-writes table of the memory system according to the third modification of the second embodiment.

[0028] FIG. 25 is a block diagram illustrating an example of an information processing system including a memory system according to a fourth modification of the second embodiment.

[0029] FIG. 26 is a conceptual diagram illustrating an example of a number-of-errors table of the memory system according to the fourth modification of the second embodiment.

[0030] FIG. 27 is a flowchart illustrating an example of a data write operation of a memory system according to a third embodiment.

DETAILED DESCRIPTION

[0031] In general, according to one embodiment, a memory system includes a semiconductor memory, a controller, and a first circuit. The semiconductor memory includes at least one memory cell configured to nonvolatily store data having at least two bits. The controller is configured to cause the semiconductor memory to execute a first write operation and a second write operation. The first write operation writes a first bit into the memory cell. The second write operation writes first data into the memory cell. The first data has at least two bits based on the first bit read from the memory cell and a second bit. The first circuit checks whether or not the first bit read from the memory cell includes a bit error. The controller is configured to cause the semiconductor memory to execute, in the second write operation, writing of the first data that includes the second bit and a third bit which is obtained by correcting the bit error of the first bit, in a case that the first bit read from the memory cell includes the bit error based on a result of checking by the first circuit.

[0032] Hereinafter, embodiments will be described with reference to the accompanying drawings. For the description, common parts are assigned common reference numerals or symbols throughout the drawings.

[0033] Each of function blocks may not be categorized as illustrated in the following example. For

example, some of functions may be implemented by a function block different from the illustrated function blocks. In addition, the illustrated function blocks may be further divided into smaller function sub-blocks. Embodiments are not limited in accordance with which function block implements the functions.

[0034] Each of the function blocks may be implemented in the form of hardware, computer software, or a combination thereof.

1. FIRST EMBODIMENT

[0035] A memory system according to a first embodiment will be described. In the following, a memory system including a semiconductor memory such as a NAND flash memory will be described as an example.

1-1. Configuration

1-1-1. Configuration of Memory System

[0036] The configuration of the memory system according to the present embodiment will be described with reference to FIG. 1. FIG. 1 is a block diagram illustrating an example of an information processing system including a memory system according to the present embodiment.

[0037] The information processing system includes a memory system **1** and a host **2**.

[0038] The memory system **1** is a device configured to store data. Examples of the memory system **1** include a solid state drive (SSD), a universal flash storage (UFS) device, a universal serial bus (USB) memory, a multi-media card (MMC), and an SD™ card. The memory system **1** can be coupled to the host **2** via a host bus. The memory system **1** performs processing based on a request signal received from the host **2** or an internal processing request.

[0039] The host **2** is a device configured to control the memory system **1**. Examples of the host **2** include a personal computer, a server system, a mobile device, a vehicle-mounted device, and a digital camera.

[0040] Next, the internal configuration of the memory system **1** will be described.

[0041] The memory system **1** includes a memory controller **10** and a semiconductor memory **30**. The semiconductor memory **30** is, for example, a nonvolatile memory such as a NAND flash memory. In the following, the semiconductor memory **30** will be referred to as a NAND flash memory **30**.

[0042] The memory controller **10** is a device configured to control the NAND flash memory **30**. The memory controller **10** is, for example, a system-on-a-chip (SoC). The memory controller **10** can be coupled to the host **2** via the host bus. The memory controller **10** receives a request signal from the host **2** via the host bus. The type of the host bus depends on an application applied to the memory system **1**. In the case of the memory system **1** being the SSD, for example, an interface under the Serial Attached SCSI (SAS), Serial ATA (SATA), or Peripheral Component Interconnect Express (PCIe™) standard is used as the host bus. In the case of the memory system **1** being the UFS device, an interface under the M-PHY standard is used as the host bus. In the case of the memory system **1** being the USB memory, an interface under the USB standard is used as the host bus. In the case of the memory system **1** being the MMC, an interface under the embedded Multi-Media Card (eMMC) standard is used as the host bus. In the case of the memory system **1** being the SD™ card, an interface under the SD™ standard is used as the host bus.

[0043] The memory controller **10** controls the NAND flash memory **30** via a NAND bus based on the request signal received from the host **2** or the internal processing request. The NAND bus performs signal transmission/reception in compliance with a NAND interface.

[0044] The NAND flash memory **30** is a device configured to store data. The NAND flash memory **30** includes a plurality of memory cell transistors. Each of the memory cell transistors nonvolatily stores data in accordance with a threshold voltage. The NAND flash memory **30** nonvolatily stores data received from the memory controller **10** in the plurality of memory cell transistors. The NAND flash memory **30** outputs data read from the plurality of memory cell transistors to the memory controller **10**.

[0045] Next, the internal configuration of the memory controller **10** will be described.

[0046] The memory controller **10** includes a host interface (I/F) circuit **11**, a processor (e.g., Central Processing Unit: CPU) **12**, a buffer memory **13**, an error checking and correcting (ECC) circuit **14**, a read-only memory (ROM) **15**, a random-access memory (RAM) **16**, and a NAND interface (I/F) circuit **17**.

[0047] The host interface circuit **11** is a circuit configured to control communications between the memory controller **10** and the host **2**. The host interface circuit **11** can be coupled to the host **2** via the host bus.

[0048] The processor **12** is a control circuit of the memory controller **10**. The processor **12** controls the overall operation of the memory controller **10** by executing a program (firmware) stored in the ROM **15**. For example, upon receipt of a write request from the host **2**, the processor **12** controls a data write operation based on the received write request. Similar processing is performed in a data read operation and a data erase operation.

[0049] The buffer memory **13** is a memory configured to temporarily store data. The buffer memory **13** is, for example, a static random-access memory (SRAM). The buffer memory **13** temporarily stores write data, read data, etc. The write data is data that is to be written into the NAND flash memory **30**. The read data is data that is read from the NAND flash memory **30**.

[0050] The ECC circuit **14** is a circuit configured to perform error checking and correcting (ECC) processing on data. Specifically, the ECC circuit **14** generates an error correction code based on write data in a data write operation. In a data read operation, the ECC circuit **14** detects an error by generating a syndrome based on the error correction code in a predetermined unit and corrects the detected error.

[0051] The ROM **15** is a nonvolatile memory. The ROM **15** is, for example, an electrically erasable programmable read-only memory (EEPROM™). The ROM **15** stores programs such as the firmware.

[0052] The RAM **16** is a volatile memory. The RAM **16** is, for example, an SRAM. The RAM **16** is used as a work area of the processor **12**. The RAM **16** stores the firmware for managing the NAND flash memory **30**, and various types of management information. The RAM **16** stores, for example, a write management table TWM and a write history table T1H. The write management table TWM and the write history table T1H will be described later in detail.

[0053] The NAND interface circuit **17** is a circuit configured to control communications between the memory controller **10** and the NAND flash memory **30**. The NAND interface circuit **17** is coupled to the NAND flash memory **30** via the NAND bus. For example, the NAND interface circuit **17** controls transfer of data, commands, addresses, etc., between the memory controller **10** and the NAND flash memory **30**.

1-1-2. Configuration of NAND Flash Memory

[0054] The configuration of the NAND flash memory **30** will be described with reference to FIG. 2. FIG. 2 is a block diagram illustrating an example of the configuration of the NAND flash memory **30**. The NAND flash memory **30** includes a memory cell array **31**, an input/output circuit **32**, a logic control circuit **33**, a ready/busy control circuit **34**, a register **35**, a sequencer **36**, a driver module **37**, a row decoder module **38**, a sense amplifier module **39**, and a data latch **41**.

[0055] The memory cell array **31** includes a plurality of blocks BLK0 to BLKn (where n is an integer equal to or larger than 1). The block BLK is, for example, a set of memory cell transistors from which data is erased collectively. The block BLK is, for example, used as a unit of a data erase operation. The block BLK includes a plurality of word lines. Each of the memory cell transistors can nonvolatily store data. A plurality of bit lines and the plurality of word lines are provided in the memory cell array **31**. Each of the memory cell transistors is associated with, for example, a single bit line and a single word line. The memory cell array **31** will be described later in detail.

[0056] The input/output circuit **32** is a circuit configured to transmit and receive signals and

information to and from the memory controller **10**. The input/output circuit **32** transmits and receives an input/output signal DQ (for example, 8-bit signals DQ0 to DQ7) and a data strobe signal DQS and a data strobe signal DOSn (an inversion signal of the signal DQS) to and from the memory controller **10**. The signal DQ is an entity of data transmitted and received between the NAND flash memory **30** and the memory controller **10**. Examples of the signal DQ include, for example, a command CMD, an address ADD, status information STS, and data DAT. The signals DOS and DOSn are signals for controlling a timing of transmitting and receiving the signal DQ. For example, when data is written, the signals DQS and DQSn along with the signal DQ including write data are transmitted from the memory controller **10** to the NAND flash memory **30**. The NAND flash memory **30** receives the signal DQ including the write data in synchronization with the signals DOS and DQSn. When data is read, the signals DOS and DOSn along with the signal DQ including read data are transmitted from the NAND flash memory **30** to the memory controller **10**. The memory controller **10** receives the signal DQ including the read data in synchronization with the signals DOS and DQSn. The input/output circuit **32** may receive the signals DOS and DQSn from the memory controller **10** via the logic control circuit **33**.

[0057] The input/output circuit **32** transmits a command CMD in the signal DQ to a command register **35A**. The input/output circuit **32** transmits an address ADD in the signal DQ to an address register **35B**. The input/output circuit **32** receives status information STS from a status register **35C**. The input/output circuit **32** transmits and receives data DAT in the signal DQ to and from the sense amplifier module **39**.

[0058] The logic control circuit **33** is a circuit configured to control the input/output circuit **32** and the sequencer **36** based on a control signal. The logic control circuit **33** receives, from the memory controller **10**, a chip enable signal CEn, a command latch enable signal CLE, an address latch enable signal ALE, a write enable signal WEn, and a read enable signal REn. The signal CEn is a signal for enabling the NAND flash memory **30**. The signal CLE is a signal indicating that the signal DQ received by the NAND flash memory **30** is a command CMD. The signal ALE is a signal indicating that the signal DQ received by the NAND flash memory **30** is an address ADD. The signal WEn is a signal instructing the NAND flash memory **30** to input the signal DQ. The signal REn is a signal instructing the NAND flash memory **30** to output the signal DQ. The NAND flash memory **30** generates the signals DQS and DOSn based on the signal REn. The NAND flash memory **30** outputs the signal DQ to the memory controller **10** based on the generated signals DQS and DQSn. The logic control circuit **33** controls the input/output circuit **32** and the sequencer **36** based on the received signal.

[0059] The ready/busy control circuit **34** is a circuit configured to notify the memory controller **10** of an operation status of the sequencer **36**. The ready/busy control circuit **34** transmits a ready/busy signal RBn to the memory controller **10** in accordance with the operation status of the sequencer **36**. The signal RBn is a signal indicative of whether the NAND flash memory **30** is in a ready state or a busy state. The signal RBn is set to a high level, for example, when the NAND flash memory **30** is in the ready state. The ready state is a state in which the NAND flash memory **30** can receive a command from the memory controller **10**. The signal RBn is set to a low level, for example, when the NAND flash memory **30** is in the busy state. The busy state is a state in which the NAND flash memory **30** cannot receive a command from the memory controller **10**.

[0060] The register **35** is a circuit configured to temporarily store information. The register **35** includes the command register **35A**, the address register **35B**, and the status register **35C**.

[0061] The command register **35A** is a circuit configured to store a command CMD. The command CMD includes, for example, an instruction to cause the sequencer **36** to execute a data read operation, a data write operation, and a data erase operation.

[0062] The address register **35B** is a circuit configured to store an address ADD. The address ADD includes, for example, a block address BAd, a page address PAd, and a column address CAd. The block address BAd, the page address PAd, and the column address CAd are used to select, for

example, a block BLK, a word line, and a bit line, respectively.

[0063] The status register **35C** is a circuit configured to temporarily store, for example, status information STS of a data read operation, a data write operation, and a data erase operation. The status information STS is used to notify the memory controller **10** of whether or not the operation has been completed normally.

[0064] The sequencer **36** is a circuit configured to control an operation of another circuit in accordance with a predetermined program. The sequencer **36** controls the overall operation of the NAND flash memory **30**. For example, the sequencer **36** controls the ready/busy control circuit **34**, the driver module **37**, the row decoder module **38**, and the sense amplifier module **39** based on a command CMD stored in the command register **35A**. For example, the sequencer **36** executes a data read operation, a data write operation, and a data erase operation.

[0065] The driver module **37** is a circuit configured to generate a voltage for use in a data read operation, a data write operation, and a data erase operation. The driver module **37** applies the generated voltage to a signal line corresponding to a selected word line based on a page address PAD stored in the address register **35B**.

[0066] The row decoder module **38** is a circuit configured to

[0067] select, based on a block address BAd stored in the address register **35B**, one of the blocks BLK in the memory cell array **31**. The row decoder module **38** transfers the voltage applied to the signal line corresponding to the selected word line to the selected word line in the selected block BLK.

[0068] In a data write operation, the sense amplifier module **39** applies a voltage which is based on write data DAT received from the input/output circuit **32** via the data latch **41**, to a bit line. In a data read operation, the sense amplifier module **39** determines data stored in a memory cell transistor based on the voltage of the bit line. The sense amplifier module **39** transfers the determination result as read data DAT to the input/output circuit **32** via the data latch **41**.

[0069] The data latch **41** includes a plurality of latch circuits (not shown). Each of the latch circuits temporarily stores write data or read data. For example, in a data write operation, the data latch **41** temporarily stores the write data received from the input/output circuit **32**, and transmits the write data to the sense amplifier module **39**. Furthermore, in a data read operation, the data latch **41** temporarily stores the read data received from the sense amplifier module **39**, and transmits the read data to the input/output circuit **32**.

1-1-3 Circuit Configuration of Memory Cell Array

[0070] The circuit configuration of the memory cell array **31** will be described with reference to FIG. **3**. FIG. **3** is a circuit diagram of the memory cell array **31**. FIG. **3** illustrates the circuit configuration of a block BLK0 included in the memory cell array **31**, as an example of the circuit configuration of the memory cell array **31**. The other blocks BLK have similar configurations to that illustrated in FIG. **3**.

[0071] The block BLK includes, for example, five string units SU0 to SU4. Each of the string units SU is, for example, a set of NAND strings NS which are selected collectively in a data write operation or a data read operation. Each of the string units SU includes a plurality of NAND strings NS respectively associated with bit lines BL0 to BLm (m is an integer equal to or larger than 1). Each of the NAND strings NS is a set of memory cell transistors coupled in series. Each of the NAND strings NS includes, for example, memory cell transistors MC0 to MC181 and select transistors ST1 and ST2. Each of the memory cell transistors MC nonvolatily stores data. Each of the memory cell transistors MC includes a control gate and a charge storage layer. The select transistors ST1 and ST2 are switching elements. Each of the select transistors ST1 and ST2 is used to select a string unit SU in various operations.

[0072] In each of the NAND strings NS, the memory cell transistors MC0 to MC181 are coupled in series. A drain of the select transistor ST1 is coupled to a bit line BL associated therewith. A source of the select transistor ST1 is coupled to one end of the memory cell transistors MC0 to MC181

coupled in series. A drain of the select transistor ST2 is coupled to the other end of the memory cell transistors MC0 to MC181 coupled in series. A source of the select transistor ST2 is coupled to a source line SL.

[0073] Control gates of the memory cell transistors MC0 to MC181 in the same block BLK are respectively coupled in common to word lines WL0 to WL181. Gates of the select transistors ST1 in the string units SU0 to SU4 are respectively coupled in common to select gate lines SGD0 to SGD4. Gates of the select transistors ST2 included in the same block BLK are coupled in common to a select gate line SGS.

[0074] In the circuit configuration of the memory cell array 31 described in the above, the bit line BL is shared by, for example, the NAND strings NS to which the same column address CAd is assigned in the respective string units SU. The source line SL is shared by, for example, the plurality of blocks BLK.

[0075] A set of memory cell transistors MC coupled to a common word line WL in one string unit SU is referred to as, for example, a cell unit CU. Each of the blocks BLK includes a plurality of cell units CU. Hereinafter, a cell unit CU, which corresponds to a pair of one of the word lines WL (word line WL_i) and one of the string units SU (string unit SU_j), will be referred to as a cell unit CU (i, j) (i is an integer equal to or larger than 0 and equal to or smaller than 181, whereas j is an integer equal to or larger than 0 and equal to or smaller than 4). In the case where the cell units CU (i, j) are not distinguished from each other, they will simply be referred to as a cell unit CU. Data stored in a cell unit CU that includes the plurality of memory cell transistors MC each storing 1-bit data in accordance with a threshold voltage is equivalent to 1-page data. The cell unit CU may store 2-page or more data based on the number of bits of data stored in the memory cell transistor MC. In the present embodiment, one memory cell transistor MC can store 4-bit data. That is, the memory cell transistor MC according to the present embodiment is a quad-level cell (QLC) configured to store 4-bit data. In this case, data stored in the cell unit CU is equivalent to 4-page data.

[0076] The number of bits of data that the memory cell transistor MC can store is not limited to 4 bits. The present embodiment is applicable to a memory cell transistor MC that can store data of multiple bits, that is, two bits or more. For example, the memory cell transistor MC may be a multi-level cell (MLC) configured to store 2-bit data, may be a triple-level cell (TLC) configured to store 3-bit data, or may be a penta-level cell (PLC) configured to store 5-bit data.

[0077] In addition, the circuit configuration of the memory cell array 31 is not limited to the configuration described in the above. For example, the number of string units SU included in each block BLK and the number of memory cell transistors MC and select transistors ST1 and ST2 included in each NAND string NS may be set freely. Hereinafter, the memory cell transistor MC will also be referred to as a memory cell MC.

1-1-4 Threshold Voltage Distributions of Memory Cells

[0078] The threshold voltage distributions to which the memory cells MC may belong will be described with reference to FIG. 4. FIG. 4 is a diagram illustrating an example of a relationship between threshold voltage distributions of the memory cells MC and data stored in the memory cells MC. The lateral axis in FIG. 4 indicates the threshold voltages of the memory cells MC. The vertical axis in FIG. 4 indicates the number of memory cells MC.

[0079] As illustrated in FIG. 4, in the case of the memory cells MC each configured to store 4-bit data, their threshold voltage distributions are divided into 16. These 16 threshold voltage distributions will be referred to as an “S0” state to an “S15” state, from lower to higher threshold voltages.

[0080] In addition, voltages V1 to V15 are voltages (hereinafter, referred to as a “verify voltage”) used to verify the “S1” state to the “S15” state, respectively, in a data write operation. The voltage VREAD is a voltage that is applied to a non-selected word line WL in a data read operation. When the voltage VREAD is applied to a gate of a memory cell MC, the memory cell MC is turned to an ON state regardless of data stored therein. A relationship between these voltages is expressed as

[0081] $V1 < V2 < V3 < V4 < V5 < V6 < V7 < V8 < V9 < V10 < V11 < V12 < V13 < V14 < V15 < VREAD$.

[0082] The “S0” state corresponds to a state in which data has been erased from the memory cell MC. Each of the “S1” state to the “S15” state corresponds to a state in which data is written by injecting electric charge into the charge storage layer of the memory cell MC. The threshold voltages of the memory cells MC included in the “S0” state are lower than the voltage V1. The threshold voltages of the memory cells MC included in the “S1” to “S14” states are equal to or higher than the voltages V1 to V14 and lower than the voltages V2 to V15, respectively. The threshold voltages of the memory cells MC included in the “S15” state are equal to or higher than the voltage V15 and smaller than the voltage VREAD.

[0083] The verify voltage and a read voltage corresponding to each of the states may have the same set value or different set values. To simplify the description, a case will be described below where a set values of the verify voltage and a set values of the read voltage are the same.

[0084] Hereinafter, data read operations using the voltages V1 to V15 will be respectively referred to as data read operations R1 to R15. The data read operation R1 determines whether or not the threshold voltage of the memory cell MC is lower than the voltage V1. The data read operation R2 determines whether or not the threshold voltage of the memory cell MC is lower than the voltage V2. The data read operation R3 determines whether or not the threshold voltage of the memory cell MC is lower than the voltage V3. The same applies to the subsequent data read operations.

[0085] As stated above, each of the memory cells MC belongs to one of the 16 threshold voltage distributions, thereby belonging to a corresponding one of the 16 states. By assigning “0000” to “1111” in binary numbers to these states, each of the memory cells MC can store 4-bit data.

Hereinafter, bits constituting 4-bit data will be referred to as a lower bit, a middle bit, an upper bit, and a top bit in order from the least significant bit. A set of lower bits stored in the memory cells MC included in the same cell unit CU will be referred to as a lower page, a set of middle bits stored in the memory cells MC included in the same cell unit CU will be referred to as a middle page, a set of upper bits stored in the memory cells MC included in the same cell unit CU will be referred to as an upper page, and a set of top bits stored in the memory cells MC included in the same cell unit CU will be referred to as a top page.

[0086] In the example illustrated in FIG. 4, with respect to the “lower bit/middle bit/upper bit/top bit” of each of the memory cells MC included in each of the states, data is allocated as described below. [0087] “S0” state: “1111” data [0088] “S1” state: “1110” data [0089] “S2” state: “1100” data [0090] “S3” state: “1101” data [0091] “S4” state: “1001” data [0092] “S5” state: “1011” data [0093] “S6” state: “0011” data [0094] “S7” state: “0010” data [0095] “S8” state: “1010” data [0096] “S9” state: “1000” data [0097] “S10” state: “0000” data [0098] “S11” state: “0001” data [0099] “S12” state: “0101” data [0100] “S13” state: “0111” data [0101] “S14” state: “0110” data [0102] “S15” state: “0100” data

[0103] Note that data allocated to the respective states is not limited to this example.

[0104] In the case of reading data allocated in this manner, the lower bit is determined through the data read operations R6, R8, and R10. The middle bit is determined through the data read operations R4 and R12. The upper bit is determined through the data read operations R2, R5, R9, R13, and R15. The top bit is determined through the data read operations R1, R3, R7, R11, and R14.

1-2 Data Write Operation

[0105] Next, the data write operation will be described. In the following, first, an overview of the data write operation will be described.

1-2-1 Overview of Data Write Operation

[0106] The data write operation includes a program operation and a verify operation. A threshold voltage of the memory cell MC is raised to a target level by repeating the program operation and the verify operation in combination.

[0107] The program operation indicates an operation of raising a threshold voltage by injecting

electrons into the charge storage layer (or maintaining a threshold voltage by inhibiting the injection). In the following, an operation of raising a threshold voltage will be referred to as a “0” program”. For example, raising the threshold voltage of a memory cell MC included in the “S0” state to be included in the “S1” state will be referred to as the “0” program. To the bit line BL which is subject to the “0” program, a voltage (for example, a voltage VSS) for the “0” program is supplied from the sense amplifier module 39. On the other hand, an operation of maintaining a threshold voltage will be referred to as a “1” program” or “write inhibit”. To the bit line BL which is subject to the “1” program, a voltage for the “1” program is supplied from the sense amplifier module 39.

[0108] The verify operation is an operation of reading data after the program operation, and determining whether or not the threshold voltage of a memory cell MC has reached a target level. Writing data to the memory cell MC whose threshold voltage has reached the target level is inhibited.

[0109] In the present embodiment, the data write operation described above is divided into two and executed separately. Specifically, the data write operation of 4-page data corresponding to a cell unit CU is divided into two: a data write operation of the lower page and the middle page (hereinafter, referred to as a “first write operation”); and a data write operation of the upper page and the top page (hereinafter, referred to as a “second write operation”), which are executed separately.

1-2-2 First Write Operation

[0110] First, the first write operation will be described

[0111] with reference to FIG. 5. FIG. 5 is a sequence diagram illustrating an example of the first write operation. In FIG. 5, the verify operation is omitted.

[0112] As illustrated in FIG. 5, the memory controller 10 causes the sequencer 36 to execute the first write operation. The sequencer 36 executes the first write operation based on the lower page data and the middle page data input from the memory controller 10 to the sense amplifier module 39. In other words, the first write operation is an operation of writing the lower bits included in the lower page data and the middle bits included in the middle page data into the respective memory cells MC within a cell unit CU.

[0113] When the first write operation is executed, the threshold voltage of the memory cell MC rises based on the write data, and four threshold voltage distributions are formed. FIG. 6 is a diagram illustrating changes in threshold voltage distributions of the memory cells MC caused by the first write operation. As illustrated in FIG. 6, an “M0” state, an “M1” state, an “M2” state, and an “M3” state are formed through the first write operation. As illustrated in FIG. 6, for example, in the first write operation, the “M0” state is formed by a plurality of memory cells MC into which “11” data is written. Herein, the expression of “kh” data indicates that the lower bit is “k” data and the middle bit is “h” data. The “M1” state is formed by a plurality of memory cells MC into which “10” data is written. The “M2” state is formed by a plurality of memory cells MC into which “00” data is written. The “M3” state is formed by a plurality of memory cells MC into which “01” data is written.

[0114] More specifically, the sequencer 36 uses voltages VM1, VM2, and VM3 as a verify voltage. The voltage VM1 is a verify voltage for use in the case of writing “10” data and is equal to or higher than the voltage V1 and lower than the voltage V4. The voltage VM2 is a verify voltage for use in the case of writing “00” data and is equal to or higher than the voltage V4 and lower than the voltage V6. The voltage VM3 is a verify voltage for use in the case of writing “01” data and is equal to or higher than the voltage V6 and lower than the voltage V12.

[0115] Threshold voltages included in the “M0” state are lower than the voltage V1. Threshold voltages included in the “M1” state are equal to or higher than the voltage VM1 and lower than the voltage V5. Threshold voltages included in the “M2” state are equal to or higher than the voltage VM2 and lower than the voltage V7. Threshold voltages included in the “M3” state are equal to or

higher than the voltage VM3 and lower than the voltage V13.

1-2-3 Second Write Operation

[0116] Next, the second write operation will be described with reference to FIG. 7. FIG. 7 is a sequence diagram illustrating an example of the second write operation. In FIG. 7, the verify operation is omitted.

[0117] As illustrated in FIG. 7, the memory controller 10 causes the sequencer 36 to execute the second write operation. The sequencer 36 executes the second write operation based on the upper page data and the top page data input from the memory controller 10 to the sense amplifier module 39, and the lower page data and the middle page data read from the memory cell array 31 to the sense amplifier module 39 and stored in the data latch 41. In other words, the second write operation is an operation of writing, into the respective memory cells MC in a cell unit CU, the lower bits included in the lower page data and the middle bits included in the middle page data read from the cell unit CU, and the upper bits included in the upper page data and the top bits included in the top page data input from the memory controller 10. The lower page data and the middle page data may be read from the memory cell array 31 to the memory controller 10. In this case, the page data received from the memory cell array 31 may be corrected in error by the memory controller 10 and the corrected page data may be supplied again as the lower page data and the middle page data from the memory controller 10 to the data latch 41.

[0118] When the second write operation is executed, the threshold voltage of the memory cell MC rises based on the write data, and sixteen threshold voltage distributions are formed. FIG. 8 is a diagram illustrating changes in threshold voltage distributions of the memory cells MC caused by the second write operation. As illustrated in FIG. 8, in the second write operation, for example, the threshold voltage distributions corresponding to the “S0” state to the “S3” state are formed from the threshold voltage distribution corresponding to the “M0” state. The threshold voltage distributions corresponding to the “S4” state, the “S5” state, the “S8” state, and the “S9” state are formed from the threshold voltage distribution corresponding to the “M1” state. The threshold voltage distributions corresponding to the “S6” state, the “S7” state, the “S10” state, and the “S11” state are formed from the threshold voltage distribution corresponding to the “M2” state. The threshold voltage distributions corresponding to the “S12” state to the “S15” state are formed from the threshold voltage distribution corresponding to the “M3” state.

1-2-4 Execution Order of Data Write Operation

[0119] Next, the order of executing the first write operation and the second write operation will be described with reference to FIG. 9. FIG. 9 is a diagram illustrating an example of the order of executing the first write operation and the second write operation on a block BLK.

[0120] FIG. 9 illustrates an example of the order of selecting the word lines WL and the string units SU in the case of continuously writing data in the block BLK0. Each of square frames in a solid line, which corresponds to intersection points of the respective word lines WL and the respective string units SU, indicates a single cell unit CU. An upper stage divided by a broken line in each square frame indicates the second write operation (reference symbol “WRT2”). A lower stage divided by the broken line in each square frame indicates the first write operation (reference symbol “WRT1”). That is, the number in the upper and lower stages in the square frame indicates the order of executing the first write operation and the second write operation (hereinafter, the order will be referred to as a “write index”) on the plurality of memory cells MC in each cell unit CU of the block BLK0.

[0121] As illustrated in FIG. 9, as the first to fifth operations, the sequencer 36 executes the first write operation in which the word line WL0 is selected and the string units SU0 to SU4 are selected in order.

[0122] Next, as the 6th operation, the sequencer 36 executes the first write operation in which the string unit SU0 is selected and the word line WL1 is selected. As the 7th operation, the sequencer 36 executes the second write operation in which the string unit SU0 is selected and the word line

WL0 is selected.

[0123] Subsequently, as the 8th to 15th operations, the sequencer **36** selects the string units SU1 to SU4 in order, and alternately executes the first write operation in which the word line WL1 is selected and the second write operation in which the word line WL0 is selected, in the same procedure as that of the 6th and 7th operations.

[0124] Next, as the 16th to 25th operations, in a similar manner to the 6th to 15th operations, the sequencer **36** selects the string units SU0 to SU4 in order, and alternately executes the first write operation in which the word line WL2 is selected and the second write operation in which the word line WL1 is selected.

[0125] Hereinafter, in a similar manner, as the $(26+10p)$ th to $(35+10p)$ th operations, the sequencer **36** selects the string units SU0 to SU4 in order, and alternately executes the first write operation in which the word line WL $(3+p)$ is selected and the second write operation in which the word line WL $(2+p)$ is selected (p is an integer equal to or larger than 0 and equal to or smaller than 178).

[0126] Next, as the 1816th to 1820th operations, the sequencer **36** executes the second write operation in which the word line WL181 is selected and the string units SU0 to SU4 are selected in order.

1-2-5 Write Management Table

[0127] The write management table TWM will be described with reference to FIG. 10. FIG. 10 is a conceptual diagram illustrating an example of the write management table TWM.

[0128] The write management table TWM stores, for each block BLK, up to which cell unit CU the data write operation (the first write operation or the second write operation) has been executed.

[0129] As illustrated in FIG. 10, each of entries of the write management table TWM includes a write destination block BLK, a cell unit CU which is a target of a data write operation (hereinafter, referred to as a “target cell unit CU”) in the write destination block BLK, and a type of the data write operation (hereinafter, referred to as a “write type”) executed on the plurality of memory cells MC in the target cell unit CU. The write destination block BLK indicates a block BLK on which the data write operation is in progress, more specifically a block BLK in which writing to all of the cell units CU has not been completed and on which the data write operation may be subsequently executed. The write type stores, for example, “1” in the case where the first write operation has already been executed, and stores “2” in the case where the second write operation has already been executed.

[0130] In the example illustrated in FIG. 10, with respect to the write destination block BLK0, the data write operation has been executed up to the second write operation on the plurality of memory cells MC in the cell unit CU (1,1). With respect to the write destination block BLK1, the data write operation has been executed up to the first write operation on the plurality of memory cells MC in the cell unit CU (0,2).

[0131] The entry is added when the first write operation is executed on the plurality of memory cells MC in the cell unit CU (0,0) of a block BLK. The entry is deleted when the second write operation is executed on the plurality of memory cells MC in the cell unit CU (181, 4) of the block BLK.

[0132] The write management table TWM is stored in one of the blocks BLK in the NAND flash memory **30**, for example. The write management table TWM is loaded from the NAND flash memory **30** to the RAM **16**, for example, immediately after the memory system **1** is powered on. Furthermore, when the memory system **1** is powered off, the write management table TWM stored in the RAM **16** is written into one of the blocks BLK in the NAND flash memory **30**.

1-2-6 Write History Table

[0133] The write history table T1H will be described with reference to FIG. 11. FIG. 11 is a conceptual diagram illustrating an example of the write history table T1H.

[0134] The write history table T1H stores time when the first write operation is executed on the plurality of memory cells MC in a target cell unit CU of a write destination block BLK. The write

history table T1H is prepared for each write destination block BLK.

[0135] As illustrated in FIG. 11, the write history table T1H has a plurality of (six in the example of FIG. 11) entries. Each of the entries includes an index, the target cell unit CU of the write destination block BLK, and time (hereinafter, referred to as “first write time”) when the first write operation was executed on the plurality of memory cells MC in the target cell unit CU. The index is, for example, 1 to 6. The first write time stores, for example, date and time information.

[0136] In the example illustrated in FIG. 11, the time when the first write operation is executed on the plurality of memory cells MC in the cell unit CU (1, 0) is 00:00:00 on Jan. 1, 2022. The time when the first write operation is executed on the plurality of memory cells MC in the cell unit CU (1,1) is 00:01:00 on Jan. 1, 2022. The time when the first write operation is executed on the plurality of memory cells MC in the cell unit CU (1,2) is 00:03:00 on Jan. 1, 2022. The time when the first write operation is executed on the plurality of memory cells MC in the cell unit CU (1, 3) is 00:10:00 on Jan. 1, 2022. The time when the first write operation is executed on the plurality of memory cells MC in the cell unit CU (1,4) is 00:10:30 on Jan. 1, 2022. The time when the first write operation is executed on the plurality of memory cells MC in the cell unit CU (2,0) is 00:11:00 on Jan. 1, 2022.

[0137] Each entry is added when the first write operation is executed on a block BLK. In the case where the first write operation is executed on the block BLK when the target cell unit CU and the first write time are stored in all of the six entries of the corresponding write history table T1H, the oldest entry is updated. In the example illustrated in FIG. 11, the oldest entry is the entry of index 1. The entry is deleted when the second write operation is executed on the plurality of memory cells MC in the target cell unit CU of the block BLK.

[0138] The write history table T1H is stored in one of the blocks BLK in the NAND flash memory 30, for example. The write history table T1H is loaded from the NAND flash memory 30 to the RAM 16, for example, immediately after the memory system 1 is powered on. Furthermore, when the memory system 1 is powered off, the write history table T1H stored in the RAM 16 is written into one of the blocks BLK in the NAND flash memory 30.

[0139] The entry in the write history table T1H may not include the target cell unit CU. In this case, the cell unit CU corresponding to the first write time in each entry in the write history table T1H is determined based on the present write index and the write management table TWM.

1-2-7 Details of Data Write Operation

[0140] Next, details of a data write operation will be described with reference to FIG. 12. FIG. 12 is a flowchart illustrating an example of the data write operation.

[0141] When the memory controller 10 receives lower page data and middle page data or upper page data and top page data from the host 2, the processing from step S100 to step S105 is executed in accordance with the writing order described above.

[0142] In the case where the first write operation is to be executed on a target cell unit CU (more specifically, the plurality of memory cells MC in a target cell unit CU) (S100_Yes), the memory controller 10 sets the target cell unit CU and the first write time to the write history table T1H (S101). The memory controller 10 sets, for example, the present time as the first write time.

[0143] Upon completion of the setting to the write history table T1H, the sequencer 36 executes the first write operation on the target cell unit CU based on the lower page data and the middle page data input from the memory controller 10 to the sense amplifier module 39 (S102).

[0144] Upon completion of the first write operation on the target cell unit CU, the memory controller 10 sets the target cell unit CU and the write type to the write management table TWM (S105). The memory controller 10 sets, for example, “1” (the first write operation) as the write type. In this manner, the memory controller 10 terminates the first write operation on the target cell unit CU.

[0145] On the other hand, in the case where the first write operation is not to be executed, that is, the second write operation is to be executed on a target cell unit CU (more specifically, the plurality

of memory cells MC in a target cell unit CU) (S100_No), the memory controller **10** deletes the entry of the target cell unit CU from the write history table T1H (S103).

[0146] When the entry of the target cell unit CU is deleted from the write history table T1H, the sequencer **36** executes the second write operation on the target cell unit CU based on the upper page data and the top page data input from the memory controller **10** to the sense amplifier module **39**, and the lower page data and the middle page data read from the memory cell array **31** to the sense amplifier module **39** and stored in the data latch **41** (S104).

[0147] Upon completion of the second write operation on the target cell unit CU, the memory controller **10** sets the target cell unit CU and the write type to the write management table TWM (S105). The memory controller **10** sets, for example, “2” (the second write operation) as the write type. In this manner, the memory controller **10** terminates the second write operation on the target cell unit CU.

1-3 Padding Operation

[0148] It is unpredictable when a write request from the host **2** will come, and the write request may not come for a long time in the course of a data write operation. A lapse of a long period of time in the course of the data write operation on the memory cell MC may lead to deterioration in reliability of data written into the memory cell MC. In order to execute the data write operation without depending on the write request from the host **2** alone and to guarantee completion of the data write operation on the memory cell MC within a fixed time period, the present operation performs a padding operation. Hereinafter, first, an overview of the padding operation will be described.

1-3-1 Overview of Padding Operation

[0149] The padding operation is an operation of filling (writing into), with some kind of data (hereinafter, referred to as “padding data”), the plurality of memory cells MC in a cell unit CU on which the first write operation has not been executed, or the plurality of memory cells MC in a cell unit CU on which the first write operation has been executed but the second write operation has not been executed. In other words, in the padding operation, the first write operation is executed, based on the padding data, on the cell unit CU on which the first write operation has not been executed. The second write operation is executed, based on the padding data, on the cell unit CU on which the first write operation has been executed but the second write operation has not been executed. The padding operation is executed in the case where a padding determination operation results in the determination that the padding operation is to be executed.

1-3-2 Padding Determination Operation

[0150] The padding determination operation will be described with reference to FIG. **13**. FIG. **13** is a flowchart illustrating an example of the padding determination operation.

[0151] The memory controller **10** executes steps S120 to S123 at fixed time intervals set in advance.

[0152] First, the memory controller **10** calculates a time difference T1 between the first write time of the oldest entry in the write history table T1H and the present time (S120). That is, the memory controller **10** calculates an elapsed time period T1 after the execution of the first write operation on the plurality of memory cells MC in a cell unit CU on which the oldest first write operation was executed.

[0153] In the case where the time difference T1 does not exceed a threshold value TVa (S121 No), the memory controller **10** terminates the padding determination operation. In this case, the memory controller **10** does not execute the padding operation. The threshold value TVa is, for example, one hour, two hours, etc.

[0154] On the other hand, in the case where the time difference T1 exceeds the threshold value TVa (S121 Yes), the memory controller **10** determines that the padding operation is to be executed, and prepares padding data (S122). Specifically, the memory controller **10** prepares, based on the write management table TWM and the write history table T1H, the padding data for use in data write

operations scheduled to be executed from (A) a data write operation next to the present write index to (B) the second write operation on the target cell unit CU of the oldest entry. In this case, the memory controller **10** executes the padding operation based on the prepared padding data (S123). The padding operation will be described later in detail.

1-3-3 Padding Data

[0155] The size of the padding data to be prepared will be described.

[0156] First, a case in which a target cell unit CU and the first write time are stored for all of the six entries in the write history table T1H will be described with reference to FIG. 14. FIG. 14 is a diagram illustrating an example of a size of padding data for use in the padding operation.

[0157] FIG. 14 describes, as an example, a case in which a data write operation has been executed up to the first write operation on the cell unit CU (2,1) (write index: 18). In this case, as illustrated in FIG. 14, the write management table TWM stores “CU (2, 1)” as the target cell unit CU of the corresponding write destination block BLK and stores “1” as the write type. In indexes 1 to 6 of the write history table T1H, “CU (2, 1)”, “CU (1, 1)”, “CU (1, 2)”, “CU (1, 3)”, “CU (1, 4)”, and “CU (2, 0)” are stored as the target cell unit CU in order from index 1, and the time when the first write operation was executed on the corresponding cell unit CU is stored as the first write time in each of these indexes. The oldest entry in the write history table T1H is the entry of index 2.

[0158] The data write operation to be executed next to the first write operation on the cell unit CU (2,1) is the second write operation on the cell unit CU (1,1) (write index: 19). The oldest entry in the write history table T1H is the entry of the cell unit CU (1,1). In this case, the target cell unit CU on which the padding operation is to be executed is the cell unit CU (1, 1). Thus, in the case where the target cell unit CU and the first write time are stored in all of the six entries in the write history table T1H, the memory controller **10** prepares the padding data (upper page data and top page data) for use in the second write operation on the target cell unit CU (cell unit CU (1,1)) of the oldest entry. That is, the size of the padding data prepared in this case is 2-page data.

[0159] Next, a case in which the write history table T1H contains an entry in which the target cell unit CU and the first write time are not stored will be described with reference to FIG. 15. FIG. 15 is a diagram illustrating an example of a size of padding data for use in the padding operation.

[0160] FIG. 15 describes, as an example, a case in which a data write operation has been executed up to the first write operation on the cell unit CU (0,2) (write index: 3). In this case, as illustrated in FIG. 15, the write management table TWM stores “CU (0, 2)” as the target cell unit CU of the corresponding write destination block BLK and stores “1” as the write type.

[0161] In indexes 1 to 3 of the write history table T1H, “CU (0, 0)”, “CU (0, 1)”, and “CU (0, 2)” are stored as the target cell unit CU in order from index 1, and the time when the first write operation was executed on the corresponding cell unit CU is stored as the first write time in each of these indexes. In indexes 4 to 6, the target cell unit CU and the first write time are not stored. The oldest entry in the write history table T1H is the entry of index 1.

[0162] The data write operation to be executed next to the first write operation on the cell unit CU (0,2) is the first write operation on the cell unit CU (0, 3) (write index: 4). The oldest entry in the write history table T1H is the entry of the cell unit CU (0,0). In this case, the target cell unit CU on which the padding operation is to be executed is the cell unit CU (0, 3), the cell unit CU (0, 4), the cell unit CU (1, 0), and the cell unit CU (0,0). Thus, in the case where the write history table T1H contains an entry in which the target cell unit CU and the first write time are not stored, the memory controller **10** prepares the padding data (lower page data and middle page data) for use in the first write operation on the cell unit CU (0, 3), the cell unit CU (0,4), and the cell unit CU (1,0). Furthermore, the memory controller **10** prepares the padding data (upper page data and top page data) for use in the second write operation on the target cell unit CU (cell unit CU (0,0)) of the oldest entry. That is, the size of the padding data prepared in this case is 2 pages×4 cell units CU=8-page data.

[0163] Next, the content of padding data to be prepared will be described.

[0164] The padding data is, for example, dummy data. The dummy data is data prepared by, for example, randomizing “0” data having the number of bits of data to be prepared, in such a manner as to prevent an imbalance between the number of 0's and the number of 1's. The randomizing is executed by, for example, the memory controller **10**. The dummy data may be, for example, random data having the number of bits of data to be prepared.

[0165] Furthermore, the padding data may be, for example, data of another block BLK. Data of said another block BLK is, for example, data to be rewritten (copied) from said another block BLK by wear leveling, compaction, or refresh.

[0166] Note that the wear leveling, compaction, and refresh are executed by the memory controller **10**.

[0167] The wear leveling is, for example, processing of leveling the number of data erase operations between the plurality of blocks in the NAND flash memory **30** by exchanging data stored in a block BLK passing through a relatively large number of data erase operations and data stored in another block BLK passing through a relatively small number of data erase operations.

[0168] The compaction is processing of reading (collecting) valid data from a plurality of blocks and rewriting the valid data in another block BLK. The compaction is performed, for example, in the case where the number of erasable blocks BLK decreases.

[0169] The refresh is processing of correcting an error in data stored in a block BLK and then rewriting the corrected data in another block BLK or the block BLK itself. The refresh is executed, for example, when the block BLK with an increased number of corrected bits is detected through the error correction processing or when a fixed time period has elapsed from the execution of a data write operation on the block BLK.

1-3-4 Details of Padding Operation

[0170] Next, details of the padding operation will be described with reference to FIG. **16**. FIG. **16** is a flowchart illustrating an example of the padding operation.

[0171] The padding operation is executed based on the elapsed time period **T1** after the execution of the first write operation on the cell unit CU on which the oldest first write operation was executed. In the case of the elapsed time period **T1** exceeding the threshold value **TVa**, that is, when the memory controller **10** prepares the padding data through the processing at step **S122** illustrated in FIG. **13** described above, the processing from step **S130** to step **S136** is executed in accordance with the write order described above. In other words, corresponding 2-page data of the prepared padding data is written in accordance with the write order into the plurality of memory cells MC in a target cell unit CU on which the padding operation is to be executed.

[0172] In the case where the first write operation is to be executed on the target cell unit CU (more specifically, the plurality of memory cells MC in the target cell unit CU) (**S130_Yes**), in the case of the padding data being the dummy data (**S131_Yes**), the sequencer **36** executes the first write operation on the target cell unit CU based on the prepared padding data (lower page data and middle page data corresponding to the target cell unit CU) input from the memory controller **10** to the sense amplifier module **39** (**S133**). In other words, the sequencer **36** writes the padding data into the target cell unit CU by executing the first write operation. On the other hand, in the case of the padding data not being the dummy data (**S131_No**), the memory controller **10** sets the target cell unit CU and the first write time to the write history table **T1H** (**S132**). Data that is not the dummy data is, for example, data that is rewritten from another block through wear leveling, compaction, or refresh. Upon completion of the setting to the write history table **T1H**, the sequencer **36** executes the processing at step **S133**.

[0173] Upon completion of the first write operation on the target cell unit CU, the memory controller **10** sets the target cell unit CU and the write type to the write management table **TWM** (**S136**). The memory controller **10** sets, for example, “1” (the first write operation) as the write type.

[0174] On the other hand, in the case where the second write operation is to be executed on the

target cell unit CU (more specifically, the plurality of memory cells MC in the target cell unit CU) (S130_No), the memory controller **10** deletes the entry of the target cell unit CU from the write history table T1H (S134).

[0175] When the entry of the target cell unit CU is deleted from the write history table T1H, the sequencer **36** executes the second write operation on the target cell unit CU based on the prepared padding data (upper page data and top page data corresponding to the target cell unit CU) input from the memory controller **10** to the sense amplifier module **39** (S135). In other words, the sequencer **36** writes the padding data into the target cell unit CU by executing the second write operation.

[0176] Upon completion of the second write operation on the target cell unit CU, the memory controller **10** sets the target cell unit CU and the write type to the write management table TWM (S136). The memory controller **10** sets, for example, “2” (the second write operation) as the write type.

[0177] Through the padding operation, the first write operation is executed on the cell units CU respectively corresponding to the write indexes using the padding data (lower page data and middle page data) from the next write index of the present write index to a write index that is one value smaller than the write index of the second write operation of the cell unit CU on which the oldest first write operation was executed. In other words, in the first write operation, the memory controller **10** causes the sequencer **36** to execute the writing of the padding data (lower page data and middle page data) into these cell units CU. Furthermore, the second write operation on the cell unit CU on which the oldest first write operation was executed is executed using the padding data (upper page data and top page data). In other words, in the second write operation, the memory controller **10** causes the sequencer **36** to execute writing of the padding data (upper page data and top page data) into the cell unit CU on which the oldest first write operation was executed. In this manner, the padding data is written into the cell units CU respectively corresponding to the write indexes from the next write index of the present write index to the write index of the second write operation of the cell unit CU on which the oldest first write operation was executed.

[0178] Note that after the memory system **1** is powered off, the memory controller **10** prepares the padding data when the memory system **1** is powered on again. Specifically, for example, the memory controller **10** prepares, based on the write management table TWM and the write history table T1H, the padding data for use in the respective data write operations from a data write operation to be executed next to the present write index to the second write operation on the target cell units CU of all entries in the write history table T1H. The memory controller **10** executes the padding operation based on the prepared padding data.

1-4 Effects

[0179] The first embodiment can improve data reliability of the memory system **1**. The effects will be described below.

[0180] In the case of writing data in the block BLK in the write order illustrated in FIG. **9** based on a write request received from the host **2**, the first write operation and the second write operation are not continuously executed in each target cell unit CU. In the example illustrated in FIG. **9**, the number of steps from the execution of the first write operation to the execution of the second write operation is 6 to 10 steps in the string units SU**0** to SU**4** of the word lines WL**0** and WL**181**, and is 11 steps in the string units SU**0** to SU**4** of the other word lines WL.

[0181] Thus, depending on the target cell unit CU, the second write operation may not be executed for a while after the execution of the first write operation on the memory cell MC, and the memory cell MC may be left for a relatively long time after the execution of the first write operation thereon. This is, for example, the case where a write request received from the host **2** is intermittent or where power is shut down and is not supplied for a while in the course of a data write operation being executed based on a write request received from the host **2**. In this case, for example, electrons may come out of the charge storage layer of the memory cell MC, and the threshold

voltage may change, thereby causing a bit error.

[0182] In the present embodiment, the memory controller **10** executes the padding determination operation at fixed time intervals set in advance. In the padding determination operation, in the case where the elapsed time period from the execution of the first write operation on the target cell unit CU of the oldest entry in the write history table T1H exceeds the threshold value, the memory controller **10** prepares padding data based on the write management table TWM and the write history table T1H. The padding data to be prepared is data used for each of data write operations scheduled to be executed from (A) a data write operation next to the present write index to (B) the second write operation on the target cell unit CU of the oldest entry. When the padding data is prepared, the memory controller **10** executes the padding operation based on the prepared padding data. In this manner, the padding data is written into the plurality of memory cells MC in each of the target cell units CU, from the target cell unit CU for a data write operation to be executed next to the present write index, to the target cell unit CU of the oldest entry.

[0183] Therefore, the configuration according to the present embodiment can prevent the memory cell MC from being left for a relatively long time after the execution of the first write operation thereon. This can reduce the possibility of a bit error occurring after the execution of the first write operation. Thus, data reliability of the memory system **1** can be improved.

2. SECOND EMBODIMENT

[0184] A second embodiment will be described. The memory system **1** according to the present embodiment differs from that of the first embodiment in terms of the write history table T1H being eliminated. In addition, the memory system **1** according to the present embodiment differs from that of the first embodiment in terms of the configuration of the NAND flash memory **30**.

Furthermore, the memory system **1** according to the present embodiment differs from that of the first embodiment in terms of a data write operation. Hereinafter, the following description will in principle focus on the features different from the first embodiment.

2-1 Configuration of Memory System

[0185] The configuration of the memory system **1** will be described with reference to FIG. **17**. FIG. **17** is a block diagram illustrating an example of the information processing system including the memory system **1**.

[0186] In the memory system **1**, the write history table T1H is eliminated. The rest of the configuration of the memory system **1** is similar to that of the first embodiment described with reference to FIG. **1**.

2-2 Configuration of NAND Flash Memory

[0187] The configuration of the NAND flash memory **30** will be described with reference to FIG. **18**. FIG. **18** is a block diagram illustrating an example of the configuration of the NAND flash memory **30**.

[0188] The NAND flash memory **30** further includes a data checking circuit **40**.

[0189] The data checking circuit **40** is a circuit configured to check whether or not the lower bit and the middle bit stored in the memory cell MC on which the first write operation has been executed have a bit error.

[0190] The data checking circuit **40** detects a bit error of data stored in the memory cell MC using an error detection code (EDC), for example. That is, checking data by the data checking circuit **40**, for example, detects the number of error bits in the lower bit and the middle bit for each cell unit CU by using the error detection code. Specifically, the sequencer **36** reads, by using the read voltages VM1, VM2, and VM3, the lower page data and the middle page data from the cell unit CU on which the first write operation has been executed. Each of the read voltages VM1, VM2, and VM3 is a voltage V_{dt} for use in verifying the lower page data and the middle page data from the memory cell array **31** in the second write operation. The read data is supplied to the data checking circuit **40**. Upon receipt of the data, the data checking circuit **40** detects the number of error bits in the read data in a predetermined unit using the error detection code. The error detection code is, for

example, a parity code, a check sum, a cyclic redundancy code (CRC), etc. Upon completion of the checking of the data, the data checking circuit **40** transmits a check result (detection value) to the memory controller **10**.

[0191] The rest of the configuration of the NAND flash memory **30** is similar to that of the first embodiment described with reference to FIG. 2.

[0192] The data checking circuit **40** may be provided in the memory controller **10**.

2-3 Data Write Operation

[0193] In the present embodiment, the data write operation includes a recovery operation. The recovery operation is an operation of correcting data of the memory cell MC for which it is determined that a bit error is present after the execution of the first write operation thereon. The recovery operation is executed based on a check result by the data checking circuit **40**. The recovery operation will be described later in detail.

[0194] The data write operation will be described with reference to FIG. 19. FIG. 19 is a flowchart illustrating an example of the data write operation.

[0195] When the memory controller **10** receives the lower page data and the middle page data or the upper page data and the top page data from the host **2**, the processing from step **S200** to step **S206** is executed in accordance with the write order described in the first embodiment.

[0196] In the case where the first write operation is to be executed on the target cell unit CU (more specifically, the plurality of memory cells MC in the target cell unit CU) (**S200_No**), the sequencer **36** executes the first write operation on the target cell unit CU based on the lower page data and the middle page data input from the memory controller **10** to the sense amplifier module **39** (**S205**).

[0197] Upon completion of the first write operation to the target cell unit CU, the memory controller **10** sets the target cell unit CU and the write type to the write management table TWM (**S206**). The memory controller **10** sets, for example, "1" (the first write operation) as the write type.

[0198] On the other hand, in the case where the second write operation is to be executed on the target cell unit CU (more specifically, the plurality of memory cells MC in the target cell unit CU) (**S200_Yes**), the sequencer **36** reads, by using the read voltages VM1, VM2, and VM3, the lower page data and the middle page data from the target cell unit CU. The read data is stored in the data latch **41** and is also supplied to the data checking circuit **40**. The data checking circuit **40** checks the lower page data and the middle page data of the target cell unit CU (**S201**). Upon completion of the checking of the data, the data checking circuit **40** transmits a check result with respect to the target cell unit CU to the memory controller **10**.

[0199] Upon receipt of the check result from the data

[0200] checking circuit **40**, the memory controller **10** determines whether or not the check result (that is, the number of error bits detected by the data checking circuit **40**) exceeds the threshold value TVb (**S202**). The threshold value TVb is, for example, equal to or larger than 1 bit and equal to or smaller than the number of error bits detectable by the data checking circuit **40**. The threshold value TVb is, for example, several bits, several tens of bits, etc.

[0201] In the case of the check result not exceeding the threshold value TVb (**S202_No**), the sequencer **36** executes the second write operation on the target cell unit CU based on the upper page data and the top page data input from the memory controller **10** to the sense amplifier module **39**, and the lower page data and the middle page data stored in the data latch **41** (**S204**).

[0202] Upon completion of the second write operation to the target cell unit CU, the memory controller **10** sets the target cell unit CU and the write type to the write management table TWM (**S206**). The memory controller **10** sets, for example, "2" (the second write operation) as the write type.

[0203] On the other hand, in the case of the check result exceeding the threshold value TVb (**S202_Yes**), that is, in the case of determination that the lower bits or the middle bits have more bit errors than the threshold value TVb, the memory controller **10** executes the recovery operation (**S203**).

2-4 Recovery Operation

[0204] The recovery operation will be described with reference to FIG. 20. FIG. 20 is a flowchart illustrating an example of the recovery operation.

[0205] Through the processing at step S202 illustrated in FIG. 19 described above, when the memory controller 10 determines that the check result exceeds the threshold value TVb, the processing from step S210 to step S220 is executed.

[0206] The memory controller 10 instructs output of the lower page data and the middle page data of the target cell unit CU stored in the data latch 41 through the processing at step S201 (S210). The memory controller 10 executes the ECC processing on the received data (S210).

[0207] In the case where the correction of the read data is successful through the processing at step S210 (S211_Yes), the lower page data and the middle page data corrected through the processing at step S210 are supplied to the NAND flash memory 30. When the NAND flash memory 30 receives the data, the sequencer 36 executes the second write operation on the target cell unit CU based on the upper page data and the top page data input from the memory controller 10 to the sense amplifier module 39, and the lower page data and the middle page data corrected through the processing at step S210 (S214). In other words, in the second write operation, the memory controller 10 causes the sequencer 36 to perform writing the lower bit and the middle bit corrected through the processing at step S210, and the upper bit and the top bit. That is, the writing is executed in such a manner that a memory cell MC storing two bits (the lower and middle bits before correction) through the first write operation stores data of four bits in total including the corrected lower and middle bits and the upper and top bits.

[0208] Upon completion of the second write operation on the target cell unit CU, the memory controller 10 sets the target cell unit CU and the write type to the write management table TWM (S215). The memory controller 10 sets, for example, “2” (the second write operation) as the write type.

[0209] On the other hand, in the case where the correction

[0210] of the read data is not successful through the processing at step S210 (S211_No), the memory controller 10 reads the lower page data and the middle page data from the target cell unit CU using voltages VM1s, VM2s, and VM3s (hereinafter, also referred to as “shifted voltages Vst”) respectively shifted from the voltages VM1, VM2, and VM3 (S212). The memory controller 10 then executes the ECC processing on the read data (S212). The amount of shift in voltage (hereinafter, referred to as a “shift amount ΔV ”) may be different for each of the read voltages, may be the same between some of the read voltages, or may be the same between all of the read voltages. Furthermore, the shift amount ΔV may be either a negative value or a positive value. The shift amount ΔV is set to an appropriate value by, for example, the memory controller 10.

[0211] In the case where the correction of the read data is successful through the processing at step S212 (S213_Yes), the lower page data and the middle page data corrected through the processing at step S212 are supplied to the NAND flash memory 30. When the NAND flash memory 30 receives the data, the sequencer 36 executes the second write operation on the target cell unit CU based on the upper page data and the top page data input from the memory controller 10 to the sense amplifier module 39, and the lower page data and the middle page data corrected through the processing at step S212 (S214).

[0212] In the case where the correction of the read data is not successful through the processing at step S212 (S213_No), it is assumed that the data of the target cell unit CU has a relatively large number of error bits. In other words, in the plurality of memory cells MC in the target cell unit CU, there may be an overlap in threshold voltage distributions between the “M0” state and the “M1” state, the “M1” state and the “M2” state, or the “M2” state and the “M3” state. When two threshold voltage distributions overlap each other, an intersection of these threshold voltage distributions indicates a position at which the number of error bits is small. To search for the intersection, the memory controller 10 executes, for example, tracking. The tracking is for searching for the

intersection (voltage) on which two threshold voltage distributions (that is, threshold voltage distributions of the “M0” and “M1” states, the “M1” and “M2” states, or the “M2” and “M3” states) of the memory cells MC overlap each other. Specifically, the memory controller **10** reads, by using read voltages VM1_t, VM2_t, and VM3_t obtained through the tracking, the lower page data and the middle page data from the target cell unit CU (S216). The memory controller **10** then executes the ECC processing on the read data (S216).

[0213] When the read data is corrected through the processing at step S216, the memory controller **10** prepares padding data (the upper page data and the top page data) for use in the second write operation on the target cell unit CU, as with the first embodiment (S217).

[0214] When the padding data is prepared, the sequencer **36** executes the second write operation on the target cell unit CU based on the padding data (the upper page data and the top page data) input from the memory controller **10** to the sense amplifier module **39** (S218). That is, in the case where the lower page data or the middle page data of the target cell unit CU have a relatively large number of error bits (for example, more than 100 error bits), the padding data, not the upper page data and the top page data, which are both to be written, is written into the target cell unit CU through the second write operation. The lower page data and the middle page data corrected

[0215] through the processing at step S216 are supplied to the NAND flash memory **30**. When the NAND flash memory **30** receives the data, the sequencer **36** writes the lower page data and the middle page data corrected through the processing at step S216 into a cell unit CU (hereinafter, referred to as a “cell unit CUa”) different from the target cell unit CU, more specifically, a plurality of memory cells in the cell unit CUa (S219).

[0216] The sequencer **36** may write the corrected lower page data and middle page data into the cell unit CUa, as its lower page data and middle page data.

[0217] The cell unit CUa may be a cell unit CU corresponding to the next write index of the present write index. For example, in the case of the target cell unit CU being the cell unit CU (1, 1) described with reference to FIG. 14, that is, the cell unit corresponding to the write index of 19, the sequencer **36** may select the cell unit CU (2,2) corresponding to the write index of 20 as the cell unit CUa. The sequencer **36** writes the corrected lower page data and middle page data into the cell unit CU (2,2) as its lower page data and middle page data.

[0218] The cell unit CUa may be a cell unit CU corresponding to a write index of a relatively large value as compared to the present write index. For example, in the case of the target cell unit CU being the cell unit CU (1,1) described with reference to FIG. 14, that is, the cell unit corresponding to the write index of 19, the sequencer **36** may select the cell unit CU (2, 4) corresponding to the write index of 24 as the cell unit CUa. The sequencer **36** writes the corrected lower page data and middle page data into the cell unit CU (2,4) as its lower page data and middle page data.

[0219] Alternatively, the sequencer **36** may write the corrected lower page data and middle page data into the cell unit CUa, as its upper page data and top page data. In this case also, the cell unit CUa may be a cell unit CU corresponding to the next write index of the present write index or may be a cell unit CU corresponding to a write index of a relatively large value as compared to the present write index.

[0220] For example, in the case of the target cell unit CU being the cell unit CU (1, 1) described with reference to FIG. 14, that is, the cell unit corresponding to the write index of 19, the sequencer **36** may select the cell unit CU (1, 2) corresponding to the write index of 21 as the cell unit CUa. The sequencer **36** writes the corrected lower page data and middle page data into the cell unit CU (1,2) as its upper page data and top page data. Alternatively, the sequencer **36** may select the cell unit CU (1, 4) corresponding to the write index of 25 as the cell unit CUa. The sequencer **36** writes the corrected lower page data and middle page data into the cell unit CU (1,4) as its upper page data and top page data.

[0221] Upon completion of writing into the cell unit CUa, the sequencer **36** writes the upper page data and the top page data input from the memory controller **10** to the sense amplifier module **39**

into a cell unit CU (hereinafter, referred to as a “cell unit CUb”) different from the target cell unit CU, more specifically, the plurality of memory cells in the cell unit CUb (S220).

[0222] The sequencer **36** may write the upper page data and top page data, which are both to be written, into the cell unit CUb, as its upper page data and top page data.

[0223] The cell unit CUb may be a cell unit CU corresponding to the next write index of the present write index. For example, in the case of the target cell unit CU being the cell unit CU (1, 1) described with reference to FIG. **14**, that is, the cell unit corresponding to the write index of 19, the sequencer **36** may select the cell unit CU (1, 2) corresponding to the write index of 21 as the cell unit CUb. The sequencer **36** writes the upper page data and the top page data, which are both to be written, into the cell unit CU (1,2), as its upper page data and top page data.

[0224] The cell unit CUb may be a cell unit CU corresponding to a write index of a relatively large value as compared to the present write index. For example, in the case of the target cell unit CU being the cell unit CU (1,1) described with reference to FIG. **14**, that is, the cell unit corresponding to the write index of **19**, the sequencer **36** may select the cell unit CU (1, 4) corresponding to the write index of **25** as the cell unit CUb. The sequencer **36** writes the upper page data and the top page data, which are both to be written, into the cell unit CU (1,4), as its upper page data and top page data.

[0225] Alternatively, the sequencer **36** may write the upper page data and the top page data, which are both to be written, into the cell unit CUb, as its lower page data and middle page data. In this case also, the cell unit CUb may be a cell unit CU corresponding to the next write index of the present write index or may be a cell unit CU corresponding to a write index of a relatively large value as compared to the present write index.

[0226] For example, in the case of the target cell unit CU being the cell unit CU (1, 1) described with reference to FIG. **14**, that is, the cell unit corresponding to the write index of 19, the sequencer **36** may select the cell unit CU (2, 2) corresponding to the write index of 20 as the cell unit CUb. The sequencer **36** writes the upper page data and the top page data, which are both to be written, into the cell unit CU (2,2), as its lower page data and middle page data. Alternatively, the sequencer **36** may select the cell unit CU (2, 4) corresponding to the write index of **24** as the cell unit CUb. The sequencer **36** writes the upper page data and the top page data, which are both to be written, into the cell unit CU (2,4), as its lower page data and middle page data.

[0227] The cell unit CUb may be the same as or different from the cell unit CUa.

2-5 Effects

[0228] As described in the first embodiment, when the memory cell MC is left for a relatively long time after the first write operation is executed thereon, a bit error in data stored in the memory cell MC may occur.

[0229] In the present embodiment, at the time of execution of the second write operation, the data checking circuit **40** executes checking on the lower page data and the middle page data of the target cell unit CU on which the first write operation has been executed. In the case of determination that the check result exceeds the threshold value TVb, the memory controller **10** executes the recovery operation.

[0230] In the recovery operation, in the case of a relatively small number of error bits, the memory controller **10** corrects the lower page data and the middle page data of the target cell unit CU. The case of the relatively small number of error bits is, for example, a case in which the lower page data and middle page data of the target cell unit CU read by using, for example, a plurality of default read voltages Vdt are corrected, or a case in which the lower page data and the middle page data of the target cell unit CU read by using a plurality of shifted voltages Vst are corrected. Upon the correction of the data, the sequencer **36** executes the second write operation on the target cell unit CU based on the upper page data and top page data input from the memory controller **10** to the sense amplifier module **39**, and the corrected lower page data and middle page data.

[0231] In the recovery operation, in the case of a relatively large number of error bits, the memory

controller **10** corrects the lower page data and the middle page data of the target cell unit CU. The case of the relatively large number of error bits is, for example, a case in which the lower page data and the middle page data of the target cell unit CU read by using a plurality of read voltages obtained through tracking are corrected. Upon the correction of the data, the corrected lower page data and middle page data are written into a cell unit CUa different from the target cell unit CU. The upper page data and the top page data input from the memory controller **10** to the sense amplifier module **39** are also written into a cell unit CUb different from the target cell unit CU. Herein, the cell units CUa and CUb may be the same or different. On the other hand, the padding data is written into the target cell unit CU. That is, the target cell unit CU does not serve as a target of a data read operation until the execution of a next data erase operation and data write operation. [0232] This can reduce the possibility that a bit error will occur. Thus, the configuration according to the present embodiment can improve data reliability of the memory system **1**.

2-6 First Modification

[0233] A memory system according to a first modification of the second embodiment will be described. The memory system **1** according to the present modification differs from that of the second embodiment in terms of the configuration of the data checking circuit **40**. The present modification is based on the precondition that data randomized to prevent an imbalance between the number of 0's and the number of 1's of 1-page data is written into a cell unit CU. Hereinafter, the following description will in principle focus on the features different from the second embodiment.

2-6-1 Data Checking Circuit

[0234] The data checking circuit **40** in the present modification counts the number of memory cells MC belonging to each of the four states (the "M0" state to the "M3" state) of the threshold voltage distributions formed by the first write operation, for example. That is, in a cell unit CU, the data checking circuit **40** counts the number of bits of the "0" data or "1" data of the lower bits and middle bits. Specifically, the sequencer **36** acquires, by using one of the read voltages VM1, VM2, and VM3, states of the plurality of memory cells MC in the cell unit CU on which the first write operation has been executed. The acquired states of the memory cells MC are supplied to the data checking circuit **40**. Upon receipt of the states of memory cells MC, the data checking circuit **40** counts the number of memory cells MC which are turned on. In other words, in the case where the states of memory cells MC in the cell unit CU on which the first write operation has been executed are acquired using the read voltage VM1, the data checking circuit **40** counts the number of memory cells MC having a threshold voltage equal to or lower than the voltage VM1.

[0235] Similarly, in the case where the states of memory cells MC are acquired using the read voltage VM2, the data checking circuit **40** counts the number of memory cells MC having a threshold voltage equal to or lower than the voltage VM2. Similarly, in the case where the states of memory cells MC are acquired using the read voltage VM3, the data checking circuit **40** counts the number of memory cells MC having a threshold voltage equal to or lower than the voltage VM3. Upon completion of counting the number of memory cells MC, the data checking circuit **40** transmits a check result (counted value) to the memory controller **10**.

[0236] The threshold value TVb for use in the processing at step S202 will be described.

[0237] Upon receipt of the check result from the data checking circuit **40** at step S202, the memory controller **10** determines whether or not the check result (the counted value) exceeds the threshold value TVb.

[0238] For example, in the case where 2-page data including the lower page data and the middle page data has a size of 32 Kbytes (KB), that is, the number of memory cells MC included in the cell unit CU is expressed as $16K \times 8$ and randomized data is written into a cell unit CU, an expected value for the number of memory cells having a threshold voltage corresponding to each of the "M0" state to the "M3" state included in the cell unit CU is expressed as $4K \times 8$. In other words, in the case where data is read using the voltage VM1 with respect to the target cell unit CU on which

the first write operation has been executed, the expected value for the number of memory cells MC having a threshold voltage corresponding to the “M0” state is expressed as $4K \times 8$. Similarly, in the case where data is read using the voltage VM2, the expected value for the number of memory cells MC having a threshold voltage corresponding to the “M0” state or the “M1” state is expressed as $4K \times 8 + 4K \times 8 = 8K \times 8$. In the case where data is read using the voltage VM3, the expected value for the number of memory cells MC having a threshold voltage corresponding to the “M0” state, the “M1” state, or the “M2” state is expressed as $4K \times 8 + 4K \times 8 + 4K \times 8 = 12K \times 8$.

[0239] The threshold value TVb is set within a range from, for example, a value smaller than the expected value to a value larger than the expected value, in consideration of the accuracy of randomization. The threshold value TVb falls within, for example, a range between the expected value -100 and the expected value $+100$. In this case, in the processing at step S202, the memory controller 10 determines whether or not the check result (the number of memory cells MC counted by the data checking circuit 40) falls within a range between the expected value -100 and the expected value $+100$. In the case where the check result falls within the aforementioned range (S202_No), the controller 10 executes the second write operation on the target cell unit CU (S204). On the other hand, in the case where the check result does not fall within the aforementioned range (S202_Yes), the memory controller 10 executes the recovery operation (S203).

2-6-2 Recovery Operation

[0240] The recovery operation will be described with reference to FIG. 21. FIG. 21 is a flowchart illustrating an example of the recovery operation.

[0241] Through the processing at step S202 illustrated in FIG. 19 described above, when the memory controller 10 determines that the check result exceeds the threshold value TVb, the processing at step S230 and steps 211 to 220 is executed.

[0242] The memory controller 10 reads the lower page data and the middle page data from the target cell unit CU using the read voltages VM1, VM2, and VM3 (S230). The memory controller 10 then executes the ECC processing on the read data (S230).

[0243] The processing from step S211 to step S220 other than step S230 is similar to that of the second embodiment described with reference to FIG. 20.

2-6-3 Effects

[0244] The present modification produces effects similar to those of the second embodiment. Further, the data checking circuit 40 according to the present modification may be implemented by a smaller circuit than the data checking circuit 40 according to the second embodiment.

2-7 Second Modification

[0245] A memory system according to a second modification of the second embodiment will be described. The memory system 1 according to the present modification differs from that of the second embodiment in terms of a data write operation. Hereinafter, the following description will in principle focus on the features different from the second embodiment.

2-7-1 Data Write Operation

[0246] In the present modification, the data write operation includes a data checking determination operation. The data checking determination operation is an operation of determining whether to execute checking on the lower page data and the middle page data of a target cell unit CU by the data checking circuit 40.

[0247] The data write operation will be described with reference to FIG. 22. FIG. 22 is a flowchart illustrating an example of the data write operation.

[0248] In the case where the second write operation is to be executed on a target cell unit CU on which the first write operation has been executed (S200_Yes), the memory controller 10 executes the data checking determination operation (S240). The data checking determination operation will be described later in detail.

[0249] In the case where the memory controller 10 determines that checking on the lower page data and the middle page data of the target cell unit CU is to be executed (S240_Yes), the data checking

circuit **40** checks the lower page data and the middle page data of the target cell unit CU (S201). [0250] On the other hand, in the case where the memory controller **10** determines that there is no need to execute checking on the lower page data and the middle page data of the target cell unit CU (S240_No), the sequencer **36** executes the second write operation on the target cell unit CU based on the upper page data and the top page data input from the memory controller **10** to the sense amplifier module **39**, and the lower page data and the middle page data read from the memory cell array **31** to the sense amplifier module **39** and stored in the data latch **41** (S204).

[0251] The processing from step S200 to step S206 other than step S240 and the processing from steps 210 to S220 are similar to those of the second embodiment described with reference to FIG. 19 and FIG. 20.

2-7-2 Data Checking Determination Operation

[0252] The memory controller **10** determines whether to execute the data checking by the data checking circuit **40**, based on an elapsed time period T2 from the execution of the first write operation on the target cell unit CU, for example. Specifically, the memory controller **10** determines whether or not the elapsed time period T2 from the execution of the first write operation on the plurality of memory cells MC in the target cell unit CU exceeds a threshold value TVc. In the case of the elapsed time period T2 exceeding the threshold value TVc, the memory controller **10** determines that the data checking by the data checking circuit **40** is to be executed. The threshold value TVc is set to, for example, an appropriate value (time period) by the memory controller **10**.

[0253] In order to calculate the time period T2, the write history table T1H described in the first embodiment may be prepared.

2-7-3 Effects

[0254] The present modification produces effects similar to those of the second embodiment. Furthermore, in the present modification, at the time

[0255] of execution of the second write operation, the memory controller **10** determines whether or not the elapsed time period T2 from the execution of the first write operation on the target cell unit CU exceeds the threshold value TVc. In the case of the elapsed time period T2 exceeding the threshold value TV, the data checking circuit **40** checks the lower page data and the middle page data of the target cell unit CU. For this reason, the data checking circuit **40** does not check data of the target cell unit CU every time the second write operation is executed. Thus, the configuration according to the present modification can suppress deterioration in the processing performance of the second write operation.

2-8 Third Modification

[0256] A memory system according to a third modification of the second embodiment will be described. The memory system **1** according to the present modification differs from that of the second embodiment in terms of containing a number-of-writes table TWN. Furthermore, the memory system **1** according to the present modification differs from that of the second modification of the second embodiment in terms of the data checking determination operation. Hereinafter, the following description will in principle focus on the features different from the second modification of the second embodiment.

2-8-1 Configuration of Memory System

[0257] The configuration of the memory system **1** will be described with reference to FIG. 23. FIG. 23 is a block diagram illustrating an example of the information processing system including the memory system **1**.

[0258] The memory system **1** further includes a number-of-writes table TWN. The number-of-writes table TWN is stored in, for example, the RAM **16**. The number-of-writes table TWN will be described later in detail. The rest of the configuration of the memory system **1** is similar to that of the second embodiment described with reference to FIG. 17.

2-8-2 Number-of-Writes Table

[0259] The number-of-writes table TWN will be described with reference to FIG. 24. FIG. 24 is a conceptual diagram illustrating an example of the number-of-writes table TWN.

[0260] The number-of-writes table TWN stores the number of times a data write operation is executed, for each block BLK.

[0261] As illustrated in FIG. 24, each of entries of the number-of-writes table TWN includes a block BLK and the number of times the data write operation is executed on the block BLK (hereinafter, referred to as “the number of writes”).

[0262] In the example illustrated in FIG. 24, the data write operation is executed **100** times for each of the blocks BLK0 and BLK1.

[0263] When a block BLK is selected as a write destination block BLK, the number of writes on the block BLK is counted up, and the counted-up number is written into the number-of-writes table TWN as the number of writes on the block BLK.

[0264] The number-of-writes table TWN is stored in one of the blocks BLK in the NAND flash memory **30**, for example. The number-of-writes table TWN is loaded from the NAND flash memory **30** to the RAM **16**, for example, immediately after the memory system **1** is powered on. When the memory system **1** is powered off, the number-of-writes table TWN stored in the RAM **16** is written in one of the blocks BLK in the NAND flash memory **30**.

2-8-3 Data Checking Determination Operation

[0265] The memory controller **10** determines whether to execute the data checking by the data checking circuit **40**, for example, based on the number of writes or the number of data erase operations with respect to the write destination block BLK (hereinafter, referred to as “the number of erasures”) including the target cell unit CU (for example, based on the number of writes stored in the number-of-writes table TWN). Specifically, the memory controller **10** determines whether or not the number of writes or the number of erasures N1 with respect to the write destination block BLK including the target cell unit CU exceeds a threshold value TVd. In the case of the number of writes or the number of erasures N1 exceeding the threshold value TVd, the memory controller **10** determines that the data checking is to be executed. The threshold value TVd is set to, for example, an appropriate value (number of times) by the memory controller **10**.

[0266] Meanwhile, wear leveling may allow all of the blocks BLK in the memory system **1** to be substantially equal to each other in the number of writes or the number of erasures N1. In such a case, for example, after detecting the fact that the number of writes or the number of erasures N1 with respect to a block BLK exceeds the threshold value TVd, the memory controller **10** may omit the data checking determination operation and determine that the data checking by the data checking circuit **40** is to be executed. Alternatively, based on the precondition that all blocks BLK are substantially equal in the number of erasures N1, the memory controller **10** may use the average number of erasures Ave N1 in a drive for a comparison with the threshold value TVd. In this case, after detecting the fact that the average number of erasures Ave_N1 exceeds the threshold value TVd, the memory controller **10** omits the data checking determination operation and executes the data checking on all blocks BLK.

2-8-4 Effects

[0267] The present modification produces effects similar to those of the second modification of the second embodiment.

2-9 Fourth Modification

[0268] A memory system according to a fourth modification of the second embodiment will be described. The memory system **1** according to the present modification differs from the second embodiment in terms of containing a number-of-errors table TEN. Furthermore, the memory system **1** according to the present modification differs from that of the second modification of the second embodiment in terms of the data checking determination operation. Hereinafter, the following description will in principle focus on the features different from the second modification of the second embodiment.

2-9-1 Configuration of Memory System

[0269] The configuration of the memory system **1** will be described with reference to FIG. **25**. FIG. **25** is a block diagram illustrating an example of the information processing system including the memory system **1**.

[0270] The memory system **1** further includes the number-of-errors table TEN. The number-of-errors table TEN will be described later in detail. The number-of-errors table TEN is stored in, for example, the RAM **16**. The rest of the configuration of the memory system **1** is similar to that of the second embodiment described with reference to FIG. **17**.

2-9-2 Number-of-Errors Table

[0271] The number-of-errors table TEN will be described with reference to FIG. **26**. FIG. **26** is a conceptional diagram illustrating an example of the number-of-errors table TEN.

[0272] The number-of-errors table TEN stores, for each block BLK, the number of times an error is detected in data read from the NAND flash memory **30** to the memory controller **10**. An error is detected through, for example, the ECC processing by the ECC circuit **14**.

[0273] As illustrated in FIG. **26**, each of entries of the number-of-errors table TEN includes a block BLK and the number of times an error is detected in the block BLK (hereinafter, referred to as “the number of error detections”).

[0274] In the example illustrated in FIG. **26**, an error is detected 10 times in the block BLK**0**. In the block BLK**1**, an error is detected 20 times.

[0275] When an error is detected in data read from a block BLK, the number of error detections on the block BLK is counted up, and the counted-up number is written into the number-of-errors table TEN as the number of error detections on the block BLK.

[0276] The number-of-errors table TEN is stored in, for example, one of the blocks BLK in the NAND flash memory **30**. The number-of-errors table TEN is loaded from the NAND flash memory **30** to the RAM **16**, for example, immediately after the memory system **1** is powered on. When the memory system **1** is powered off, the number-of-errors table TEN stored in the RAM **16** is written in one of the blocks BLK in the NAND flash memory **30**.

2-9-3 Data Checking Determination Operation

[0277] The memory controller **10** determines whether to execute the data checking by the data checking circuit **40**, based on a past error history of the write destination block BLK including the target cell unit CU (for example, based on the number of error detections stored in the number-of-errors table TEN). Specifically, the memory controller **10** determines whether or not the number of past error detections N2 of the write destination block BLK including the target cell unit CU exceeds a threshold value TVe. In the case of the number of error detections N2 exceeding the threshold value TVe, the memory controller **10** determines that the data checking is to be executed. The threshold value TVe is set to, for example, an appropriate value (number of times) by the memory controller **10**. Note that, the present modification described the number-of-errors table TEN storing the number of error detections for each block BLK; however, the number-of-errors table TEN may store the number of error detections for each word line WL in the block BLK. In this case, the data checking determination operation is performed based on a past error history for each word line WL in the block BLK.

[0278] The memory controller **10** may determine whether to execute the data checking by the data checking circuit **40**, based on a correction capability of the ECC circuit **14**, for example. In this case, in the case where an indicator indicating the correction capability of the ECC circuit **14** configured to perform the ECC processing on data of a target cell unit CU exceeds a threshold value TVf, the memory controller **10** determines that the data checking is to be executed. The threshold value TVf is set to, for example, an appropriate value by the memory controller **10**.

[0279] The memory controller **10** may determine that the data checking by the data checking circuit **40** is to be executed, by using the past error history in combination with the correction capacity of the ECC circuit **14**.

2-9-4 Effects

[0280] The present modification produces effects similar to those of the second modification of the second embodiment.

3. THIRD EMBODIMENT

[0281] A third embodiment will be described. The memory system **1** according to the present embodiment differs from that of the second embodiment in terms of a data write operation. The present embodiment will describe the data write operation in the case where there are a relatively large number of error bits in the lower page data or the middle page data of a cell unit CU on which the first write operation has been executed. As an example, the present embodiment performs the second write operation based on the upper page data and the top page data after execution of pre-processing of the second write operation based on padding data, with respect to the target cell unit CU on which the first write operation has been executed. Hereinafter, the following description will in principle focus on the features different from the second embodiment.

3-1 Data Write Operation

[0282] The data write operation will be described with reference to FIG. 27. FIG. 27 is a flowchart illustrating an example of the data write operation.

[0283] When the memory controller **10** receives the lower page

[0284] data and the middle page data or the upper page data and the top page data from the host **2**, the processing from step S300 to S308 is executed in accordance with the write order described in the first embodiment.

[0285] Steps S300, S301, S306, S307, and S308 are the same as steps S200, S201, S204, S205, and S206 illustrated in FIG. 19 described in the second embodiment, respectively.

[0286] Upon receipt of a check result through the processing at step S301 from the data checking circuit **40**, the memory controller **10** determines whether or not the check result (that is, the number of error bits detected by the data checking circuit **40**) exceeds a threshold value TVg (S302). The threshold value TVg is, for example, 100 bits.

[0287] In the case of the check result not exceeding the threshold value TVg (S302_No), the sequencer **36** executes the second write operation on the target cell unit CU based on the upper page data and the top page data input from the memory controller **10** to the sense amplifier module **39**, and the lower page data and the middle page data read from the memory cell array **31** to the sense amplifier module **39** and stored in the data latch **41** (S306).

[0288] On the other hand, in the case of the check result exceeding the threshold value TVg, that is, in the case of a relatively large number of error bits (S302 Yes), the memory controller **10** prepares padding data for use in the pre-processing of the second write operation on the target cell unit CU (S303).

[0289] For example, the padding data is data obtained by setting all pieces of the lower page data of the target cell unit CU to “0” data and all pieces of the middle page data of the target cell unit CU to “1” data. Each piece of the padding data used herein is not randomized data but data indicative of all “0” data or all “1” data when it is written into the target cell unit CU. In other words, out of the lower page data, the middle page data, the upper page data, and the top page data which are written into the target cell unit CU, the present embodiment replaces the lower page data and the middle page data having a relatively large number of error bits with the padding data.

[0290] When the padding data is prepared, the sequencer **36** executes the pre-processing of the second write operation on the target cell unit CU based on the padding data (the lower page data and the middle page data) input from the memory controller **10** to the sense amplifier module **39** (S304). The pre-processing of the second write operation is performed by the sequencer **36** executing the data write operation to cause threshold voltages of the respective memory cells MC in the target cell unit CU to transition to the “M3” state. Remember that all of the memory cells MC in the target cell unit CU after the first write operation exhibit threshold voltages equal to or lower than the “M3” state. Thus, the sequencer **36** can cause the threshold voltages of the

respective memory cells MC in the target cell unit CU to transition to the “M3” state by performing the pre-processing of the second write operation. The pre-processing of the second write operation may be implemented by the first write operation.

[0291] Upon completion of the pre-processing of the second write operation on the target cell unit CU, the sequencer **36** executes the second write operation on the target cell unit CU based on the upper page data and the top page data input from the memory controller **10** to the sense amplifier module **39**, and the padding data (the lower page data and the middle page data) (**S305**). That is, the sequencer **36** executes the data write operation to cause threshold voltages of the respective memory cells MC in the target cell unit CU to transition to one of the “S12” state to the “S15” state.

[0292] Note that, in the case where the number of error bits exceeds a predetermined number as a result of checking by the data checking circuit **40**, the memory controller **10** may designate a write destination block BLK to be written, as a bad block (unwritable block BLK).

[0293] Furthermore, before the processing at step **S301**, the memory controller **10** may execute the data checking determination operation described in the second modification to the fourth modification of the second embodiment.

3-2 Effects

[0294] As described in the first embodiment, when the memory cell MC is left for a relatively long time after the first write operation is executed thereon, a bit error in data stored in the memory cell MC may occur.

[0295] In the present embodiment, at the time of execution of the second write operation, the data check circuit **40** executes checking on the lower page data and the middle page data of the target cell unit CU on which the first write operation has been executed. In the case of the check result exceeding the threshold value TVg, that is, in the case of a relatively large number of error bits, the sequencer **36** executes the data write operation to cause threshold voltages of the respective memory cells MC in the target cell unit CU to transition to one of the “S12” state to the “S15” state. In this manner, the data write operation can be executed on the target cell unit CU based on valid data (i.e., the upper page data and the top page data).

4. MODIFICATIONS, ETC.

[0296] As described above, a memory system (**1**) according to an embodiment includes a semiconductor memory (**30**), a controller (**10**), and a first circuit (**40**). The semiconductor memory (**30**) includes at least one memory cell (MC) configured to nonvolatily store data having at least two bits. The controller (**10**) is configured to cause the semiconductor memory (**30**) to execute a first write operation and a second write operation. The write operation writes a first bit (lower/middle) into the memory cell (MC). The second write operation writes first data (lower/middle/upper/top) into the memory cell (MC). The first data has at least two bits based on the first bit (lower/middle) read from the memory cell (MC) and a second bit (upper/top). The first circuit (**40**) checks whether or not the first bit (lower/middle) read from the memory cell (MC) includes a bit error. The controller (**10**) is configured to cause the semiconductor memory (**30**) to execute, in the second write operation, writing of the first data (lower/middle/upper/top) that includes the second bit and a third bit (lower/middle) which is obtained by correcting the bit error of the first bit (lower/middle), in a case that the first bit (lower/middle) read from the memory cell includes the bit error based on a result of checking by the first circuit (**40**).

[0297] The embodiments are not limited to those described in the above, and various modifications may be made.

[0298] The order of the steps in the flowcharts described in the above embodiments may be altered to the extent possible.

[0299] Furthermore, the data checking circuit **40** according to the second modification to the fourth modification of the second embodiment and the third embodiment may be the circuit described in the first modification of the second embodiment.

[0300] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A memory system comprising: a semiconductor memory including a memory cell configured to nonvolatily store data having at least two bits; and a controller configured to cause the semiconductor memory to execute: a first write operation of writing first data into the memory cell; and a second write operation of writing second data into the memory cell, the second data having at least two bits that include a first bit and a second bit, wherein the controller is further configured to: determine, based on an elapsed time period from execution of the first write operation, whether to receive third data read from the memory cell on which the first write operation of writing the first data has been executed, in executing the second write operation on the memory cell; and in a case where the controller determines to receive the third data read from the memory cell, receive the third data from the semiconductor memory; obtain the first bit based on the third data read from the memory cell; transfer the second bit to the semiconductor memory; and cause the semiconductor memory to execute, in the second write operation, writing of the second data that includes the obtained first bit and the transferred second bit.
2. The memory system according to claim 1, wherein the semiconductor memory further includes a data latch, wherein the controller is further configured to, in a case where the controller determines not to receive the third data read from the memory cell, transfer the second bit to the semiconductor memory; and cause the semiconductor memory to: store, in the data latch, the third data read from the memory cell; obtain the first bit based on the third data read from the memory cell and stored in the data latch; and execute, in the second write operation, writing of the second data that includes the obtained first bit and the transferred second bit.
3. The memory system according to claim 1, wherein the controller includes a first circuit configured to check whether or not the third data read from the memory cell includes a bit error, wherein the controller is further configured to, in a case where the controller determines to receive the third data read from the memory cell, check, by using the first circuit, whether or not the third data read from the memory cell includes the bit error; and in a case where the third data read from the memory cell includes the bit error based on a result of checking by the first circuit, obtain the first bit by correcting the bit error of the third data.
4. The memory system according to claim 3, wherein the controller is further configured to, in a case where the first circuit fails to correct the bit error of the third data, cause the semiconductor memory to change a read voltage to read the third data from the memory cell.
5. The memory system according to claim 1, wherein the controller is configured to determine to receive the third data read from the memory cell, in a case where the elapsed time period from the execution of the first write operation is longer than a threshold.
6. The memory system according to claim 1, further comprising: a first circuit configured to check whether or not the third data read from the memory cell includes a bit error, wherein the controller is further configured to determine whether to execute the checking by the first circuit, in executing the second write operation on the memory cell on which the first write operation has been executed.
7. The memory system according to claim 6, wherein the controller is configured to determine whether to execute the checking by the first circuit, based on the elapsed time period from the

execution of the first write operation.

8. The memory system according to claim 7, wherein the controller is configured to determine to execute the checking by the first circuit, in a case where the elapsed time period from the execution of the first write operation is longer than a threshold.

9. The memory system according to claim 6, wherein the controller is further configured to, in a case where the first circuit fails to correct the bit error of the third data, cause the semiconductor memory to change a read voltage to read the third data from the memory cell.

10. The memory system according to claim 1, wherein the memory cell is configured to nonvolatily store the data in accordance with a threshold voltage, the first data corresponds to a first threshold voltage, and the second data, which includes the first bit and the second bit, corresponds to a second threshold voltage higher than the first threshold voltage.

11. A method of controlling a semiconductor memory that includes a memory cell configured to nonvolatily store data having at least two bits, the method comprising: causing the semiconductor memory to execute a first write operation of writing first data into the memory cell; and causing the semiconductor memory to execute a second write operation of writing second data into the memory cell, the second data having at least two bits that include a first bit and a second bit, wherein the method further comprises: determining, based on an elapsed time period from execution of the first write operation, whether to receive third data read from the memory cell on which the first write operation of writing the first data has been executed, in executing the second write operation on the memory cell.

12. The method of according to claim 11, further comprising: determining to receive the third data read from the memory cell; and in response to determining to receive the third data read from the memory cell, receiving the third data from the semiconductor memory; obtaining the first bit based on the third data read from the memory cell; transferring the second bit to the semiconductor memory; and causing the semiconductor memory to execute, in the second write operation, writing of the second data that includes the obtained first bit and the transferred second bit.

13. The method according to claim 11, wherein the semiconductor memory further includes a data latch, wherein the method further comprises: determining not to receive the third data read from the memory cell; and in response to determining not to receive the third data read from the memory cell, transferring the second bit to the semiconductor memory; causing the semiconductor memory to store, in the data latch, the third data read from the memory cell; causing the semiconductor memory to obtain the first bit based on the third data read from the memory cell and stored in the data latch; and causing the semiconductor memory to execute, in the second write operation, writing of the second data that includes the obtained first bit and the transferred second bit.

14. The method according to claim 11, further comprising: managing a first circuit configured to check whether or not the third data read from the memory cell includes a bit error; determining to receive the third data read from the memory cell; and in response to determining to receive the third data read from the memory cell, checking, by using the first circuit, whether or not the third data read from the memory cell includes the bit error; determining that the third data read from the memory cell includes the bit error based on a result of checking by the first circuit; and in response to determining that the third data read from the memory cell includes the bit error, obtaining the first bit by correcting the bit error of the third data.

15. The method according to claim 14, further comprising: determining that the first circuit fails to correct the bit error of the third data; and in response to determining that the first circuit fails to correct the bit error of the third data, causing the semiconductor memory to change a read voltage to read the third data from the memory cell.

16. The method according to claim 11, further comprising: determining that the elapsed time period from the execution of the first write operation is longer than a threshold; and in response to determining that the elapsed time period from the execution of the first write operation is longer than the threshold, determining to receive the third data read from the memory cell.

- 17.** The method according to claim 11, further comprising: managing a first circuit configured to check whether or not the third data read from the memory cell includes a bit error; and determining whether to execute the checking by the first circuit, in executing the second write operation on the memory cell on which the first write operation has been executed.
- 18.** The method according to claim 17, further comprising: determining that the elapsed time period from the execution of the first write operation is longer than a threshold; and in response to determining that the elapsed time period from the execution of the first write operation is longer than the threshold, determining to execute the checking by the first circuit.
- 19.** The method according to claim 17, further comprising: determining that the first circuit fails to correct the bit error of the third data; and in response to determining that the first circuit fails to correct the bit error of the third data, causing the semiconductor memory to change a read voltage to read the third data from the memory cell.
- 20.** The method according to claim 11, wherein the memory cell is configured to nonvolatily store the data in accordance with a threshold voltage, the first data corresponds to a first threshold voltage, and the second data, which includes the first bit and the second bit, corresponds to a second threshold voltage higher than the first threshold voltage.
-